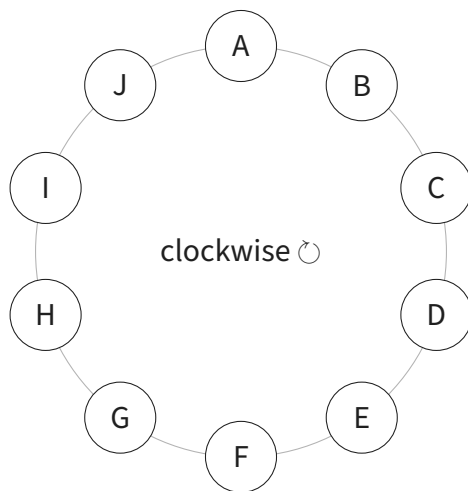


Problem 1: Use a counter on this ring. A turn moves a fixed number of places clockwise; after J comes A. Try a two-place turn from A, then from I. Record the two landings. Count places moved, not the starting place.

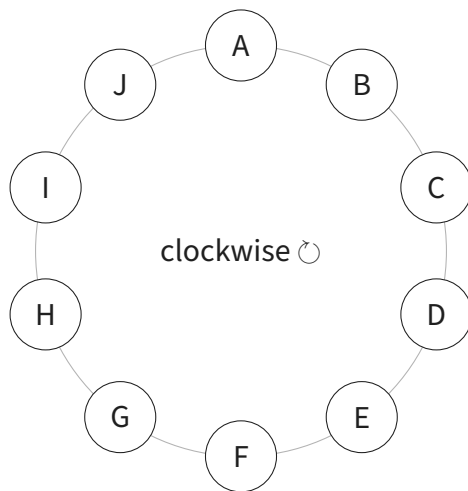


Problem 2: A secret turn takes A to D. Find its step size. Use that same turn from B, I, and J. Record the three outputs.

Problem 3: Encode BAG by shifting each letter with the secret turn. Decode GDG by undoing that turn. Check both messages by reversing your moves.

Problem 4: Someone says B becomes F with that same key. Test the claim on the ring. Record what B actually becomes.

Problem 5: Start at A. Take two clockwise places every turn. Record each landing until A first returns. Repeat from B. Do not record letters you pass over.



Start A: A →

Start B: B →

Problem 6: Try the two-place rule from C, then from D. Compare with your first two routes. Circle which route each new start joins.

Start	Joins A's route	Joins B's route
C	yes / no	yes / no
D	yes / no	yes / no

Problem 7: Mark the A-route letters with a dot and the B-route letters with a small cross on the ring. Follow one more turn from each letter and compare its marking with the next landing.

Problem 8: Use the page-2 ring. From A, take three places per turn and record the complete route until A first returns. Compare its visited letters with the two-place routes.

Problem 9: Test five-place turns from A, then from B. Draw the two routes. Compare their lengths with the two-place and three-place routes.

Problem 10: A two-place route skips letters. Test whether you can still undo a two-place move from A, B, I, and J. Find one clockwise step size that brings every test back. Record each out-and-back trip.

Problem 11: Choose a new step size from 1 to 9. Predict whether it visits all ten letters from A. Test and record its route.